6.5.3 Mutation theory

Hugo de Vries put forth the Mutation theory. Mutations are sudden random changes that occur in an organism that is not heritable. De Vries carried out his experiments in the Evening Primrose plant (*Oenothera lamarckiana*) and observed variations in them due to mutation.

According to de Vries, sudden and large variations were responsible for the origin of new species whereas Lamarck and Darwin believed in gradual accumulation of all variations as the causative factors in the origin of new species.

Hugo de Vries believed that Mutations are random and directionless, but Darwinian variations are small and directional.

Hugo de Vries believed that speciation are due to Mutation and called saltation (single step large Mutation).

Salient features of Mutation Theory

- Mutations or discontinuous variation are transmitted to other generations.
- In naturally breeding populations, mutations occur from time to time.
- There are no intermediate forms, as they are fully fledged.
- They are strictly subjected to natural selection.

6.5.4 Modern synthetic theory

Sewell Wright, Fisher, Mayer, Huxley, Dobzhansky, Simpson and Haeckel explained Natural Selection in the light of Post-Darwinian discoveries. According to this theory gene mutations, chromosomal mutations, genetic recombinations, natural selection and reproductive isolation are the five basic factors involved in the process of organic evolution.

i. **Gene mutation** refers to the changes in the structure of the gene. It is also called gene/ point mutation. It alters the phenotype of an organism and produces variations in their offspring.

ii. **Chromosomal mutation** refers to the changes in the structure of chromosomes due to deletion, addition, duplication, inversion or translocation. This too alters the phenotype of an organism and produces variations in their offspring.

iii. **Genetic recombination** is due to crossing over of genes during meiosis. This brings about genetic variations in the individuals of the same species and leads to heritable variations.

iv. **Natural selection** does not produce any genetic variations but once such variations occur it favours some genetic changes while rejecting others (driving force of evolution).

v. **Reproductive isolation** helps in preventing interbreeding between related organisms.

6.5.5 Evolution by anthropogenic sources

Natural Selection (Industrial melanism)

Natural selection can be explained clearly through industrial melanism. Industrial melanism is a classical case of Natural selection exhibited by the peppered moth, *Biston betularia*. These were available in two forms, white winged and black winged. Before industrialization peppered moth both white and black coloured were common in England. Pre-industrialization witnessed more white winged moths than dark winged moths because of the dense growth of white coloured lichens on trees. They escaped from predators due to the white background. Post-industrialization caused the tree trunks became dark due to smoke and soot let out from the industries. The lichen population also decreased because they could not grow in polluted areas. The white winged moths were easily identified by their predators as the background turned black due to soot and smoke while the black winged moths camouflaged with the dark background. Hence the dark coloured moth population was selected and their number increased when compared to the white moths. Nature

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